Metabolomics Data Matrix

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Many variables in many samples

As in all *omic* experiments, a metabolomic assay measures a large number of **variables** (properties) in a (hopefully) reasonably large number of **samples**.

Question

... what are the variables ...

1. In a targeted metabolomics investigation?

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2. In an untargeted MS based metabolomics analysis?

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1. In a targeted metabolomics investigation?
2. In an untargeted MS based metabolomics analysis?
3. In an untargeted NMR based metabolomics assay?

Number of variables | place your guess

- Targeted metabolomics investigation

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- Targeted metabolomics investigation
  - from 10 to 300

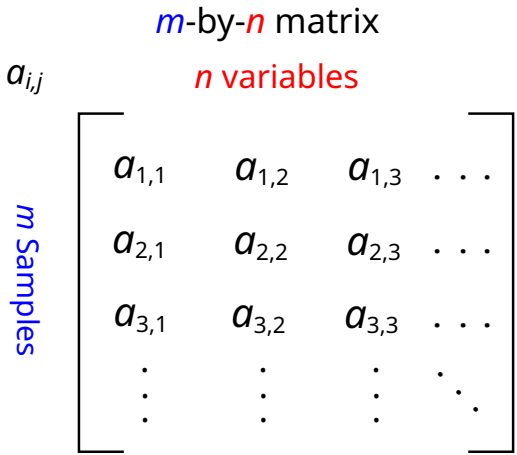
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Number of variables | place your guess

- Targeted metabolomics investigation
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- Untargeted metabolomics investigation
  - from 1000 to 30000

Data Matrix


m-by-n matrix
$a_{i,j}$
n variables
m Samples
$a_{1,1}$ $a_{1,2}$ $a_{1,3}$ . . .
$a_{2,1}$ $a_{2,2}$ $a_{2,3}$ . . .
$a_{3,1}$ $a_{3,2}$ $a_{3,3}$ . . .

The goal of data analysis

Goal

The objective of data analysis is to find **"organization"** inside the matrix

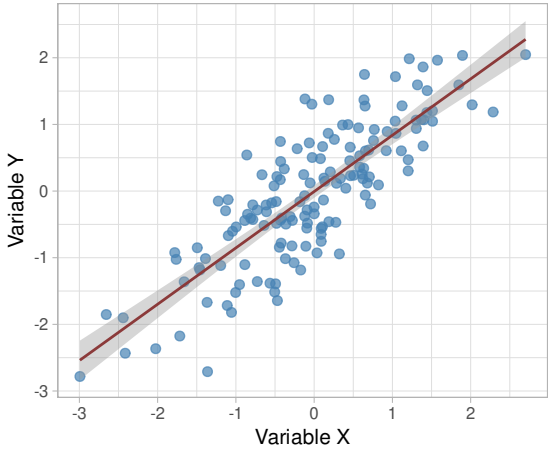
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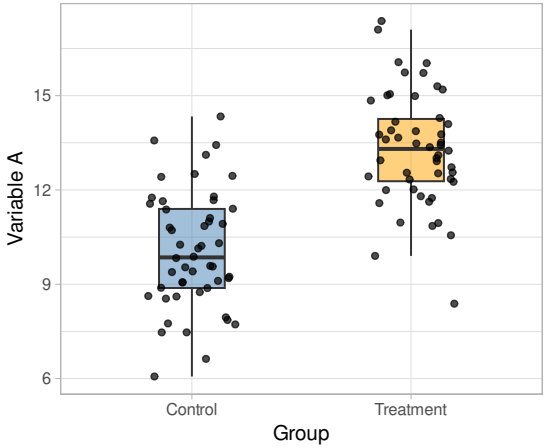
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Our organization should also be statistically robust: if someone redid my experiment, the results should be the same

Organization: Correlations



Organization: Biomarkers

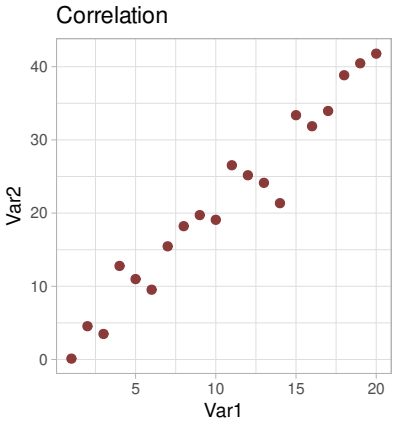


Section 1

Data Matrices: facts

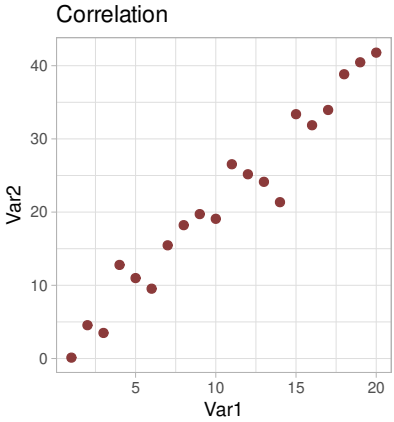
Variables are not independent

- Biological relations



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- Biological relations
- Chemical/Analytical reasons



Analytical vs biological correlation

Important

Analytical correlation is always stronger than the biological one!

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The correlation structure of your matrix is often not really informative about biology

Rows (samples) are often dependent

The design of the study can result in a multilevel hierarchical structure of the samples which violates independence. In other words some samples are "by construction" associated and share something in common

Examples

1. Repeated measures of the same individual

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2. Subpopulations

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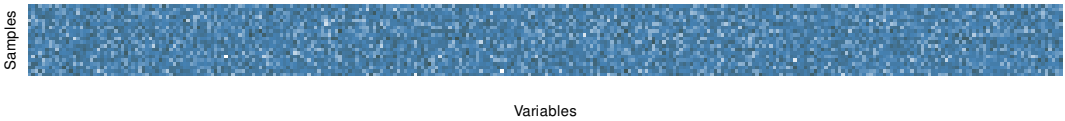
Examples

1. Repeated measures of the same individual
2. Subpopulations
3. Different sites, different days, ...

Data Matrix size

In the typical "happy" statistical context, the number of variables we measure is smaller than the number of samples.

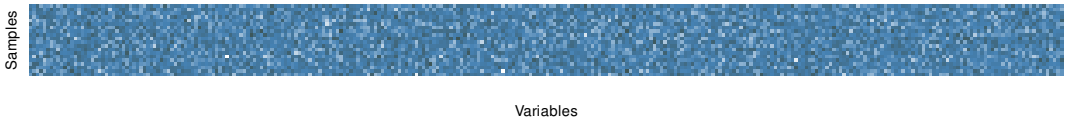
In *omics* the number of variables normally exceeds the number of samples



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FAT DATA MATRIX

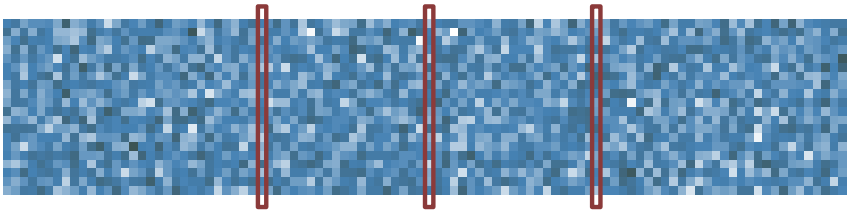
Section 2

Analyzing data matrices

Univariate approach

The **Univariate** approach considers each variable separately and applies "standard" statistical tools to spot the more **interesting** variables

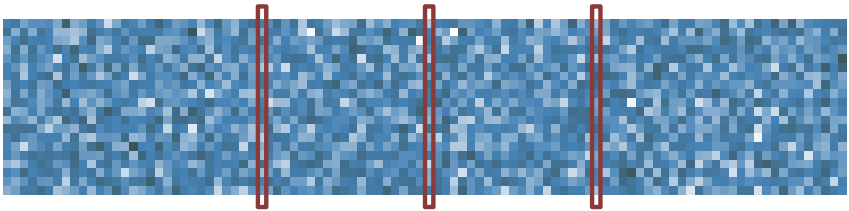
1. Statistical testing



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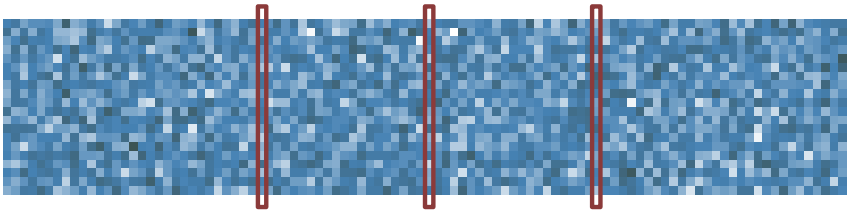
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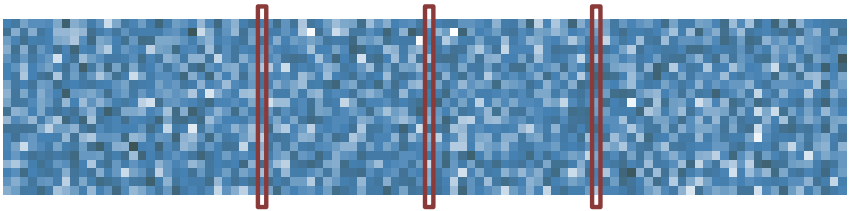
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3. ANOVAs



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1. Statistical testing
2. Linear modeling (`lm`, `glm`, ...)
3. ANOVAs
4. Hierarchical Modeling



Univariate: pros and cons

PRO

1. Statistical modeling is there!
2. Interpretable by construction

Univariate: pros and cons

PRO

1. Statistical modeling is there!
2. Interpretable by construction

CONS

1. Multiple testing
2. The structure of the data creates redundancy
3. Assumptions for parametric (and non parametric) approaches are often not fulfilled

Where is the problem: Multiple testing

- We measure many variables (features, metabolites) in the same set of samples

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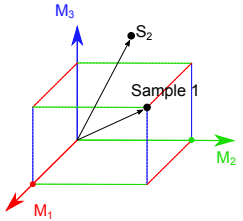
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- We run a battery of statistical tests looking for the significance of what we see in the individual variables
- **We ask ourselves if at least one test is significant in the overall set of tests**

We run individual tests, but we have a question about the full set of variables ...

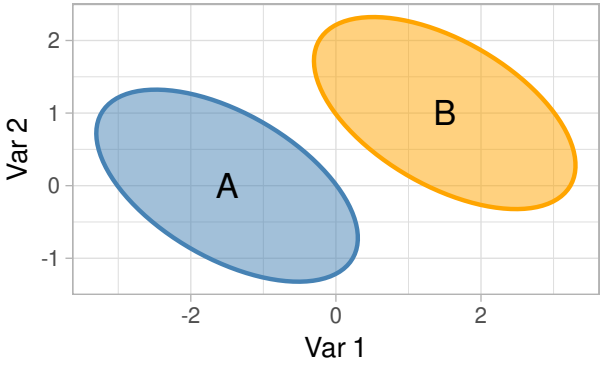
Multivariate approach

Each sample is represented as a point in the multidimensional variable space. The dataset is a **cloud** of points in that space.

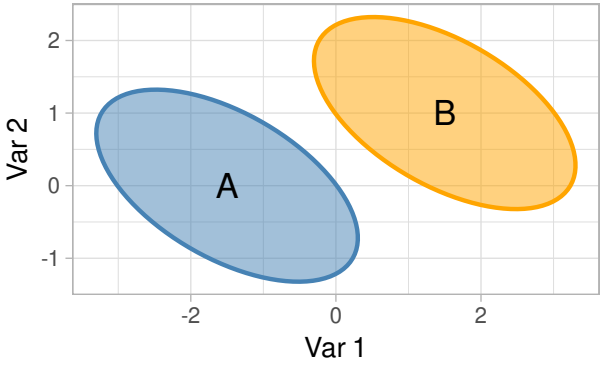
The size of the space equals the number of variables we are measuring. **Multivariate methods** (PCA, PLS, ASCA, ...) are able to exploit the correlation between the variables to highlight/extract the organization of the data



Why multivariate ...



Why multivariate ...



Group separation is clearer in the 2D space ...

Multivariate: pros and cons

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1. Potentially more powerful
2. Explicit use of variable correlation
3. No issues with multiple testing

Multivariate: pros and cons

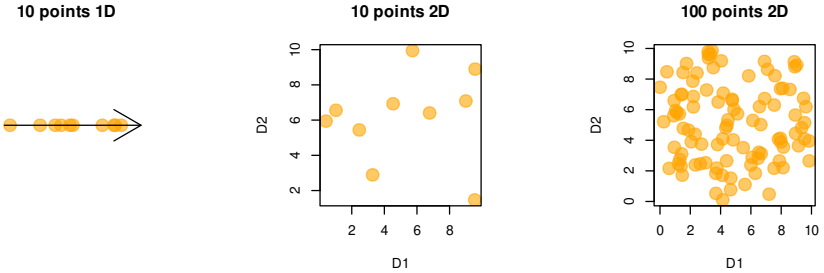
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CONS

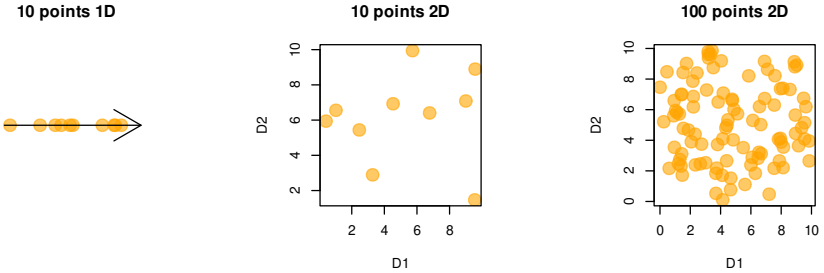
1. Chance correlations in fat matrices
2. Empty space
3. Difficult to embed hierarchical structure

The problem: the curse of dimensionality



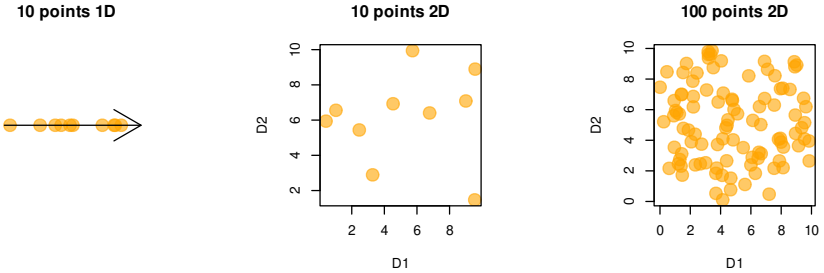
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The problem: the curse of dimensionality



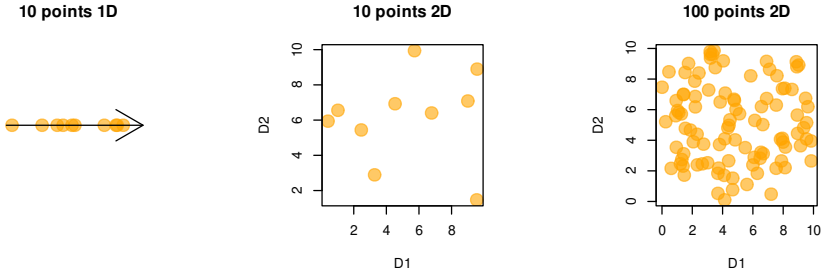
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Can you imagine 20 samples in 10000 dimensions?